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SI 507

MSI Course Networks

Introduction

For my final project in SI 507, I have written a program that helps UMSI master's students on the Big Data Analytics and UX tracks plan out and visualize their course schedules. This is a project I've been working on since SI 506, starting with simple functions and coding techniques, and it is something I have enjoyed building on my own time. SI 507 has enabled me to further develop this project with graph theory and object oriented programming.

The Data

I am using the official course catalog for UMSI courses (link below), provided by the school for students to plan out their upcoming semester. This data source is stored in a Google Sheets page, and the upper-level courses have their own tab within the sheet. I then downloaded the tab as a csv file. Accessing the data for this project was simple, while processing it was the more challenging task.

There are 119 classes listed in my data, and 102 of them are accessible to graduate students, so I was able to remove some. After clearing the unnecessary columns and renaming the remaining columns, I was left with "catalog_number", "course_title", "prerequisites", "course_description", and "credits". I then extracted the number of each course listed in the prerequisites column using regular expressions and pandas methods, resulting in a list of numbers. The problem with this column is that the catalog overlists prerequisites. What I mean by this is that a high-level course, such as SI 671, lists 507, 618, and 544 as prerequisites, while

https://docs.google.com/spreadsheets/d/1FiG9NnyZGJdMLeuY7ZliRMbN4LPDrx_8ZUlyRMaAT_Q/edit?gid=2027687471#gid=2027687471

544 is already a prerequisite of 618. This created an unnecessary, and deceptive, edge in my network. I want my network to show the flow of courses, so I needed to reduce some prerequisites. At first I tried to do this programmatically, but I, nor UMGPT, could figure it out. I ended up brute force cleaning the data, which wasn't too bad as there are only 102 courses in my data, and only a proportion of them have prerequisites. Once I cleaned my data, the course network could be built.

The Functions

My course network program first creates a directed graph of all graduate courses. I then prompt the user with five options. They can build and visualize a schedule based on their track and selective requirements, find the most central course of their schedule or all classes, find the shortest path between courses, find the prerequisites for each course, and view course information. This program has a command line interface that is primarily dictated by “\n” symbols in print statements and `time.sleep()` functions in my code. The interface of this project is something I hope to improve in the future.

Interesting Highlights

The directed graph visualization of a track-specific course schedule is useful. Sometimes it's easy to forget that our classes are connected and leading us somewhere, and the graph helps remind us of this. The visual also makes it clear which courses are not very connected (SI 500). Another highlight is that SI 507 has the best betweenness centrality score, suggesting that it is the most central course to Big Data Analytics and the MSI program, at least according to my directed graph.

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MSI BDA Course Visual

